7.5 Factoring $x^2 + bx + c$

Name _____ Period ____ Date _____

Algebra I 7.5 (30-minute plan) · Student sheet

I can…

- factor $x^2 + bx + c$ when c is positive.
- factor $x^2 + bx + c$ when c is negative.
- solve equations and real-life problems by factoring.

Vocabulary · fill in the missing word

Factoring $x^2 + bx + c$ — _____ it as $(x + p)(x + q)$, where $p + q = b$ and $pq = c$

Sign rules — $c > 0$: p and q have the same sign as b. $c < 0$: p and q have _____ signs; the larger one has the sign of b

Part 1 · Notes and examples (fill in each note, then work the example)

Factoring $x^2 + bx + c$: $x^2 + bx + c = (x + p)(x + q)$, where $p \cdot q =$ _____ and $p + q =$ _____.

Sign rules: $c > 0$: When c is positive, p and q have the _____ sign, the sign of _____.

Check: Always check by _____.

Example 1

Factor $x^2 + 7x + 10$.

Sign rules: $c < 0$: When c is negative, p and q have _____ signs; the one with the larger absolute value has the sign of _____.

Example 2

Factor $x^2 + 3x - 28$.

Part 2 · Partner practice

Partner practice

Factor $x^2 - 11x + 18$.